

Mohammad Rafieian

(945) 527-5229 | mohammad.rafieian@utdallas.edu | mohpydev.github.io | **LinkedIn:** mohammad-rafieian | **GitHub:** mohPYdev

SUMMARY

Full-Stack Software Engineer and Ph.D. Researcher with expertise in building scalable, secure systems. Experienced in architecting robust backend infrastructure and Dockerized deployment. Deep technical knowledge in Program Analysis, Software Security, and applying Machine Learning for automated vulnerability detection by Static analysis, Fuzzing and Symbolic execution.

EDUCATION

University of Texas at Dallas, M.S & Ph.D. in Software Engineering and Security, Advisor: Dr. Shiyi Wei Expected Dec 2026
Isfahan University, B.S in Software Engineering, ACM Student Chapter member Aug 2019 – Jun 2023

PUBLICATIONS

- “Is Call Graph Pruning Really Effective? An Empirical Re-evaluation”, **M.Rafieian**, et al., ICSE 2026.
- “An Empirical Study of Static Analysis-Based Variability Bug Detection”, A.Mordahl, **M.Rafieian**, et al., ISSTA 2026 (in submission).
- “Improving ML-based Static Analysis Classification via Explainable AI”, S.Yerramreddy, **M.Rafieian**, et al., ICST 2026 (under review).
- “Automatic Test Suites for Static Analysis Tools via Dynamic Analysis”, A.Mordahl, **M.Rafieian**, et al., ISSTA 2026 (in submission)

WORK EXPERIENCE

Research Assistant | University of Texas at Dallas Aug 2023 – Present

- Led research in program analysis and **security**, applying ML to enhance automated vulnerability detection for C/Java systems
- Designed large-scale experimental pipelines, using **fuzzing** and dynamic instrumentation to benchmark **4 static analysis** tools.
- Authored **4 papers**, introducing **3 novel datasets** and methodologies in static analysis research.

Backend Developer | EMK Corp Feb 2023 – Jul 2023

- Executed the backend system using the **MVC** design pattern and created the database schema for **Postgres** database.
- Developed **Django** REST API backend supporting **10k+** users with **Redis** caching, , cutting API response time by **40%**.
- Deployed the **Docker** containers to **Linux** http servers with **99.9%** uptime.

PROJECTS

Call Graph Pruning Framework | python, java, bash Feb 2025 – Jul 2025

- Built static call graphs (DOOP, OPAL, WALA) and **instrumented** Java bytecodes(ASM) to capture execution traces at runtime.
- Implemented neural networks with **PyTorch** and **TensorFlow**, automating hyperparameter tuning and fold splitting pipelines.
- Fine-tuned **CodeBERT/CodeT5** models, refining attention mechanisms to improve call graph edge classification accuracy by **30%**.
- Re-evaluated pruning methods, fixing dataset flaws; showed results inflated **(+50% vs. actual +5%)**.

Explainable AI for Static Analysis | python, java, c, bash Jan 2025 – Sep 2025

- Constructed the first explainable dataset for static analysis ML (1,132 **Java** samples, 67K **C** samples) evaluating **7 code LLMs**.
- Improved model interpretability with **Explanation-Guided Learning**, boosting relevant line focus from **35% to 65%**.
- Enhanced model classification accuracy with **Input×Gradient** xAI method by **20%**.

Fuzzing for Static Analysis Benchmarking | python, Java, c Jul 2025 – Present

- Built a novel fuzzing pipeline to auto-generate benchmarks for static analysis, reducing manual work by **80 person-hours**.
- Engineered a hybrid fuzzing approach (**directed, grey box**) with a novel feedback mechanism for critical points in static call graphs.

ArtCom | python, JavaScript, bash Feb 2022 – Jun 2022

- Developed an online art auction and exhibition platform using **React.js** and Django REST Framework reducing renting costs by **90%**.
- Implemented Dockerized architecture with **WebSocket-based** real-time auctioning supporting **200** concurrent participants.
- Designed PostgreSQL + Redis backend with user management, authentication, and **data synchronization**, easy deployment on Vercel.

Lord | python, JavaScript, SQL Aug 2022 – Dec 2022

- Pioneered a low-code reservation system generator that automated website creation, which cut deployment costs by over **90%**.
- Engineered a scalable full-stack solution (Django/**MySQL**/React) with integrated payment processing for seamless deployment.

SKILLS

Programming Languages:	Python, Java, C, C++, JavaScript, Scala, SQL
Frameworks & Libraries:	Django, Flask, PyTorch, TensorFlow, Scikit-learn, Node.js, React
Development Tools:	Git, Docker, Linux, Firebase, Vercel, PostgreSQL, Redis
Specialized Areas:	Program Analysis, Automated Software Testing, ML Pipelines, Backend Systems